01 - Nuclear Forces

Nuclear Physics PHYS2829

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Summary

The nucleus is made of protons and neutrons. The stability of the nucleus relies on the short range nuclear force. The models produced can account for the nature and energy of radiation products, the results of scattering experiments, and the range of isotopes across the table of elements. We get a measure of stability, the *binding energy per nucleon* that peaks around iron.
The natural distance scale for the nucleus is the femtometre (1 fm = $10^{-15} m$), the natural energy or mass unit is the mega electron volt ($1 MeV = 1.6 \times 10^{-13} J = 1.78 \times 10^{-30} kg$)

Contents

Nuclear Constituents

Nuclei consist of *protons* and *neutrons*

these have almost exactly the same mass

neutrons a little heavier, but by less than 0.2%

masses of nuclei are very nearly integers (when measured in amu)

intrinsic spin (and thus magnetic moment) of nucleus consistent with both neutrons and protons have spin $\frac{1}{2}$

Notation

ⓘ Syntax

- Atomic Number (**Z**) is the number of protons in the nucleus.
- Atomic Mass (**A**) is the combined number of protons and neutrons in the nucleus.
- Also, (**N**) is the number of neutrons, but we use that less often

$^{A}_{Z}Element\ e.g.\ ^{4}_{2}He\ has\ A = 4\ and\ Z = 2\ (sometimes\ just\ ^{4}He)$

how many protons, neutrons in $^{226}_{88}Ra$? How about ^{56}Fe ?

Isotopes are nuclei with the same number of protons but different numbers of neutrons

e.g. $^{208}_{82}Pb, ^{207}_{82}Pb, ^{206}_{82}Pb, ^{204}_{82}Pb$

Name	Symbol	Mass in amu	Rest Energy (MeV)	Spin
Proton	p	1.007276	938.28	½
Neuton	n	1.008665	939.57	½

A solitary neutron isn't stable, decays with half-life of 878s (15 minutes)

turns in to a proton and electron (and anti-neutrino)

Neutrons and protons aren't point particles (unlike the electron). Have radii of about 0.8fm.

made of *quarks*

$$up^{+\frac{2}{3}} \quad up^{+\frac{2}{3}} \quad down^{-\frac{1}{3}} = proton^{+1}$$

$$up^{+\frac{2}{3}} \quad down^{-\frac{1}{3}} \quad down^{-\frac{1}{3}} = neutron^{0}$$

Measuring mass in mega-electron volts (MeV)

- electron volts measure energy
  - literally energy an electron gets when it crosses a 1 volt battery
  - charge on electron is $1.6 \times 10^{-19} C$, so energy is $1eV = 1.6 \times 10^{-19} Joules$
- from relativity, $mass \leftrightarrow energy$
- using $E = m \times c^2$ thus $mass = energy \,/\, c^2$ and we get:
- $1eV = 1.6 \times 10^{-19} / \, c^2 = 1.6 \times 10^{-19} / \, (3 \times 10^8)^2 = 1.78 \times 10^{-36} kg$
- $1MeV = 10^6 eV = 1.78 \times 10^{-30} kg$

Makes for a handy unit to measure nuclear energies and masses.

Nuclear Size

femtometer is natural unit for nucleus

$1fm \ = \ 10^{-15}m$

nuclear radii range from 1fm to about 7fm

very small compared to size of atom

measure using:

electron diffraction

isotope wavelength shifts

Coulomb energies of mirror nuclei

scattering of α particles

α decay lifetimes

Electron Diffraction

electrons have wavelengths given by:

$$\lambda = \frac{hc}{E}$$

diffraction minima at

$$sin\theta = \frac{1.22 \times \lambda}{D}$$

420MeV ($\lambda = 2.95fm$) electrons give minimum at 50.5° for $^{12}_{6}C$
gives nuclear radius of 2.33fm

Isotope Wavelength Shifts

From Krane - Introductory Nuclear Physics
The larger the nucleus, the less chance the electron (or in diagram above, muon) has to be in close proximity to a high concentration of charge

Mirror Nuclei

mirror nuclei have opposite numbers of protons and neutrons

example ${}^3_1H$ and ${}^3_2He$ or ${}^7_3Li$ and ${}^7_4Be$

as far as the nuclear force is concerned, there is no difference between protons and neutrons

but the extra positive charge in ${}^3_2He$ carries a Coulomb penalty, which will depend on the average separation of the protons

can measure this energy difference by the maximum energy of positron from β decay

α Decay

looking within a nucleus, we see temporary assemblages of α particles.

these aren't static, and collide up against the inner wall of the nucleus frequently

classically they don't have enough energy to escape, but can get out through quantum mechanical *tunnelling*

super sensitive to size of barrier, and the energy of this quasi-$alpha$ particle

explains the extreme range of α half lives

get correlation between half life and ejected α energy

So, what is the size of a nucleus?

nuclear densities pretty much all the same

$Radius \ = \ 1.2 \times \sqrt[3]{A}$ in fm

surface a little proton rich

but charge and matter radii of nuclei agree to within 0.1fm

shape pretty spherical, a little oblate. Sometimes.

Structure of Nucleus

must be something holding nucleus together, stopping positive protons flying apart

also need to explain radiation

energies of $\alpha,\ \beta,\ \gamma$

nuclear half lives

scattering experiments (fire particles at nucleus and see how they bounce off)

which nuclei are stable

behind all this is the ***Nuclear Force***, the left over remnants of the ***Strong Force***

Nuclear Force

Something has to hold the nucleus together. Some properties of this *nuclear force*

must be big, has to overcome Coulomb repulsion between protons

$E_{pp} = \frac{1}{4\pi\epsilon_\circ}\frac{e^2}{r}$ - this is about 1MeV for ~fm separation

in fact, quantum mechanics says it must be even bigger

particle in a box of size of a = 6fm (something like aluminium)

$E = \frac{3\hbar^2\pi^2}{2m_pa^2} \approx 20MeV$

Nuclear Force

only relevant at fm distances, feeble at greater separations (>3fm)

see this from proton scattering

only effects protons and neutrons, not electrons

it’s like electrons are nuclear force *neutral*

agnostic to difference between protons and neutrons

repulsive at short (< 0.5fm) distances

depends on spins of nucleons

Nuclear Force

no analytic form of nuclear force

nothing like Coulomb's Law or Einstein's gravity

$F_{Coulomb} \; = \; \frac{1}{4\pi\epsilon_\circ} \; \frac{q_1 q_2}{r^2}$ or $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$

but it does have an exponential term

$F_{nuclear} \propto e^{-\frac{r}{R_0}}$

best modelled as an *exchange* force, mediated by some particle

first analysed by Yukawa

Yukawa Exchange Force

fire a proton at a neutron

it will scatter off in different directions

have to obey conservation of energy and momentum

most will go straight through pretty much

a neutron is a small target

intensity decreases with angle

but too many are scattered at up to 180°

explained by the proton becoming the neutron and vice-versa has to be some particle being passed from neutron $\rightleftharpoons$ proton

only possible if we *borrow* energy using the uncertainty principle

$\Delta E \Delta t \geq \frac{\hbar}{2}$

Δt given by time taken to cross nucleus at speed of light

$\Delta t \leq \frac{R}{c} = \frac{3 \times 10^{-15}}{3 \times 10^{8}} = 10^{-23} s$

this gives $\Delta E \approx \frac{\hbar c}{2 \times R}$

gives a particle of about 100MeV (c.f. proton 938MeV, electron 0.511MeV)

eventually discovered the **Pion** ($\pi^{\circ}\ and\ \pi^{\pm}$), which is made up a quark and anti–quark and has a mass of about 140MeV.

Investigating Nuclear Force - Deuteron

can get a bound state of a single proton and a single neutron, called *deuteron* (c.f. deuterium, ^2H)

something like the equivalent of a hydrogen atom for atomic structure

the simplest possible system

don't get *proton-proton* or *neutron-neutron*

deuteron has the protons and neutron spins parallel ($l = 1$, triplet)

no excited states of deuteron (unfortunately)

Investigating Nuclear Force - Deuteron

depth of potential well is about 35MeV (remember, $E = \frac{3\hbar^2\pi^2}{2m_p a^2}$, for deuteron we measure radius, = a/2, to be 2.1fm)

but separation energy (like *ionisation energy*) is 2.224MeV

get this from binding energies and also from formation and disassociation energies (e.g. $^1H + n \rightarrow {}^2H + \gamma$)

so deuteron *only* just exists, luckily for us

Nuclear Shell Model

Next level of complexity moves beyond simple potential well

protons and neutrons moving in orbits within nucleus

energy levels akin to atomic electron energy levels

nuclei with evens numbers of protons and even numbers of neutrons most stable

protons and neutrons fill up energy levels independently

of the 288 nuclei found in nature, 169 are even-even, only 8 are odd-odd

$^{2}D, ^{8}Li, ^{10}B, ^{14}N, ^{40}K, ^{50}V, ^{138}La, ^{176}Lu$

Nuclear Shell - Quantum States

would like to be able to solve the Schrödinger equation for the nuclear potential, just like we can to solve the electronic levels of Hydrogen

not possible because don't have a clean equation for the nuclear force/potential

but can use some approximations to get useful results

end up with a ladder of quantum levels (c.f. 1s, 2s, 2p… atomic orbitals)

transitions between levels lead to the emission of light - γ rays

separate suites of levels for protons and for neutrons

Mateusz Sitarz - 2019 thesis - nuclear energy levels for $^{116}_{50}Sn$

electrostatic repulsion between protons isn't nothing

they're only ~ 1fm apart

preference for neutrons (no repulsion)

light nuclei have Z ~ N (equal numbers)

heavier nuclei have more neutrons than protons

set of magic numbers (a.k.a. noble elements)

2, 8, 20, 28, 50, 82, 126

lots of isotopes for these elements

Nuclear Valley of Stability (from wikipedia)

Nuclear Binding Energy

nucleus is a little lighter than the sum of its parts

ⓘ Binding Energy per Nucleon of ^{238}U

- mass of proton = 938.272 MeV
- mass of neutron = 939.565 MeV
- mass of ^{238}U = 221 743 MeV

$$(938.272 \times 92 + 939.565 \times 146 - 221743)/238 = 7.372\ MeV\ per\ nucleon$$

measure of how stable a nucleus is

want this as big as possible

can repeat this calculation for all nuclei

B/A rises rapidly through the lightest elements

hits a maximum of ~8.75 MeV around ^{56}Fe

(well, actually ^{62}Ni)

decreases gradually beyond that

if we can turn elements into ^{56}Fe, from either direction, we can release energy

fission splits up heavier elements, moves up curve from the right

fusion coalesces lighter elements, moves up curve from the left

Semiempirical Mass Formula

made of five terms

an positive volume term proportional to A given by $a_\nu \ A$

a negative surface term given by $-a_s A^{-\frac{2}{3}}$

remember $Radius \ \propto \ A^{-\frac{1}{3}}$

a negative Coulomb term depending on the number and separation of the protons. Given by $-a_c Z(Z-1)A^{-\frac{1}{3}}$ (again, remember the radius)

a negative symmetric term favouring equal numbers of protons and neutrons, especially for light nuclei. Given by $-a_{sym}\frac{(A-2Z)^2}{A}$

a positive pairing term favourising even numbers of protons / neutrons. Let’s call this term just δ

Binding Energy Equation

$B = a_\nu A - a_s A^{-\frac{2}{3}} - a_c Z(Z-1)A^{-\frac{1}{3}} - a_{sym}\frac{(A-2Z)^2}{A} + \delta$

$\delta = \pm a_p A^{-\frac{3}{4}}$ “+” for Z/N even/even “-” for Z/N odd/odd else 0

a good fit for:

$a_\nu = 15.5MeV$

$a_s = 16.8MeV$

$a_c = 0.72MeV$

$a_{sym} = 23MeV$

$a_p = 34MeV$

Vol. + Surf + Coulomb + Symm + Pairing

Equations

$1MeV = 1.6 \times 10^{-13}J = 1.78 \times 10^{-30}kg$

$Radius\ = \ 1.2 \times \sqrt[3]{A}$ in femto-metres (1 fm = $10^{-15}m$)

$Binding\ E.\ per\ nucleon = \frac{938.272 \times Z + 939.565 \times (A - Z) - (mass\ in\ MeV)}{A}$

$B = a_\nu A - a_s A^{-\frac{2}{3}} - a_c Z(Z-1)A^{-\frac{1}{3}} - a_{sym}\frac{(A-2Z)^2}{A} + \delta$

References

Young & Freedman - chapter 43.2 (Nuclear Structure & Binding)

Serway & Jewett - chapter 43.3 (Nuclear Models)

Serway & Jewett - chapter 43.2 (Binding Energy)

nucleides

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Nuclear Physics

Speaker notes